



Machine Learning

CASE STUDY 28: Retail Store Sales Prediction

1. Title

Retail Store Sales Prediction Using Machine Learning

2. Problem Statement

A retail chain wants to predict the monthly sales of a store based on attributes such as store size, location type, number of staff, promotional spend, competitor distance and month of the year. Accurate prediction would support inventory and staffing decisions. Dataset: BigMart Sales Data (Kaggle) is the closest public match, although its attributes differ from those listed. Otherwise, students may generate synthetic data with the same attributes using NumPy/Pandas or `make_regression()` from scikit-learn, and must explain how it was generated.

3. Objectives

Students should:

- Perform exploratory data analysis and visualisation.
- Analyse historical store sales records.
- Handle missing values in competitor distance and promotional data.
- Encode categorical attributes such as store type and location.
- Study the effect of promotional spending on sales.
- Develop and train the specified regression models.
- Compare model performance using appropriate metrics and k-fold cross-validation.
- Perform residual analysis on the best model.
- Select the best-performing model with justification.
- Deploy the model as an interactive prediction application.
- Analyse limitations and real-world applicability.

4. Machine Learning Algorithms

- Linear Regression
- Polynomial Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor



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5. Comparative Study

Compare:

- R-squared (R^2)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Residual Plot

Students must particularly analyse whether the relationship between promotional spend and sales is linear, since promotional returns typically saturate beyond a certain expenditure level.

6. Deployment

Develop a web application where the user enters store attributes and receives:

- Predicted Monthly Sales
- Expected error range

The application should also display the selected model and prediction details where appropriate.

7. Final Analysis

Students should determine:

- Factors most influential in store sales.
- Best-performing algorithm.
- Whether promotional spending shows saturating returns.
- Whether the model supports inventory planning.
- Limitations of predicting sales without external economic data.

8. Questions to Be Answered Using ML

- Can store sales be predicted from store attributes?
- Which factor most influences sales?
- Does promotional spending show diminishing returns?
- Does polynomial regression improve the fit?
- Which algorithm achieves the highest R^2 ?
- How well does the model perform on a new store?
- Can the model be deployed for inventory planning?